Control Systems

BRANCH: ELECTRICAL (EE)
COURSE CODE: EE302

Subject Overview & Scope

This comprehensive document serves as the official compiled study notes for Control Systems (EE302) under the Electrical (EE) department. It is structured into four core university syllabus units covering primary concepts, definitions, formulas, and advanced technical applications for university examinations and placements.

University Course Syllabus & Core Units

Unit 1: Mathematical Modeling

Open loop vs Closed loop control systems. Transfer functions, block diagram reduction, Mason's Gain Formula for Signal Flow Graphs. Feedback characteristics.

Unit 2: Time Response Analysis

Standard test signals, Response of First and Second order systems. Steady-state errors and static error coefficients. PID Controllers.

Unit 3: Stability & Root Locus

Concept of stability. Routh-Hurwitz stability criterion. Root Locus technique: construction rules, angle of departure, breakaway points.

Unit 4: Frequency Response Analysis

Bode plots, Gain margin, Phase margin. Nyquist stability criterion, polar plots. State-space analysis representation.

Key Formulas & Exam Pointers

CRITICAL FORMULA / PRINCIPLE:

*Mason's Gain Formula: T = Sum(Pk * Delta_k) / Delta. Standard 2nd order characteristic eq: s^2 + 2*zeta*wn*s + wn^2 = 0*